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Department of CSE

B.Tech. CSE Trimester -5

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Alluru Amulya

Cloud Infrastructure and Services

25CS2206E

LAB RECORD

2026

Experiment -1

Experiment -1-b

AIM: Enabling Multi-Factor Authentication to Secure Your Access

Description:

Multi-Factor Authentication (MFA) is a security mechanism that provides an additional layer of protection to an AWS account during the login process. It requires users to verify their identity using two or more authentication factors instead of relying only on a username and password.

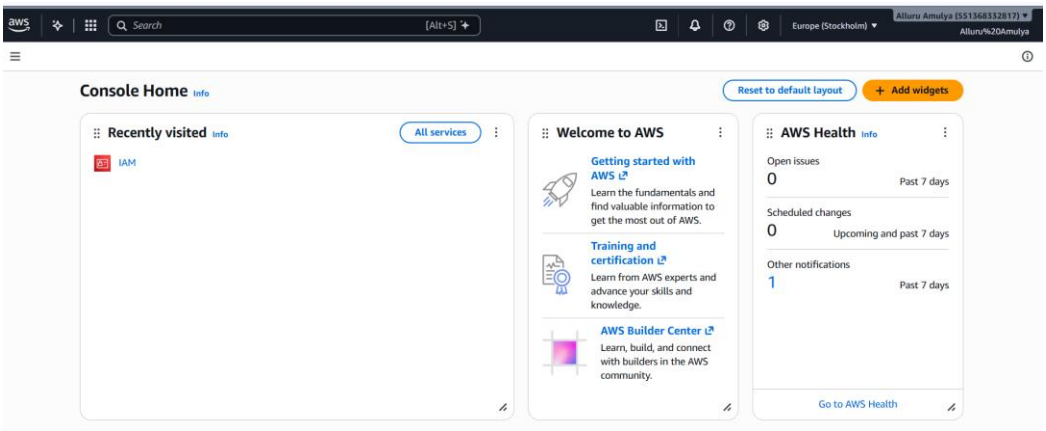
In AWS, MFA helps prevent unauthorized access even if the user's password is compromised. It can be enabled for IAM users using different authentication methods, such as an authenticator application, passkey, or security key.

In this experiment, an IAM user is created with appropriate permissions, and Multi-Factor Authentication is enabled to strengthen account security. The configured MFA device is then tested by signing out and logging in again using the IAM username, password, and additional authentication verification.

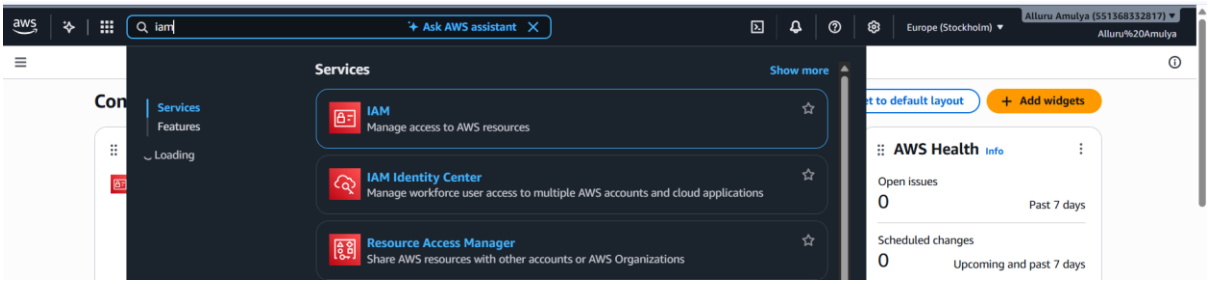
Steps:

Part A: Creating an IAM User

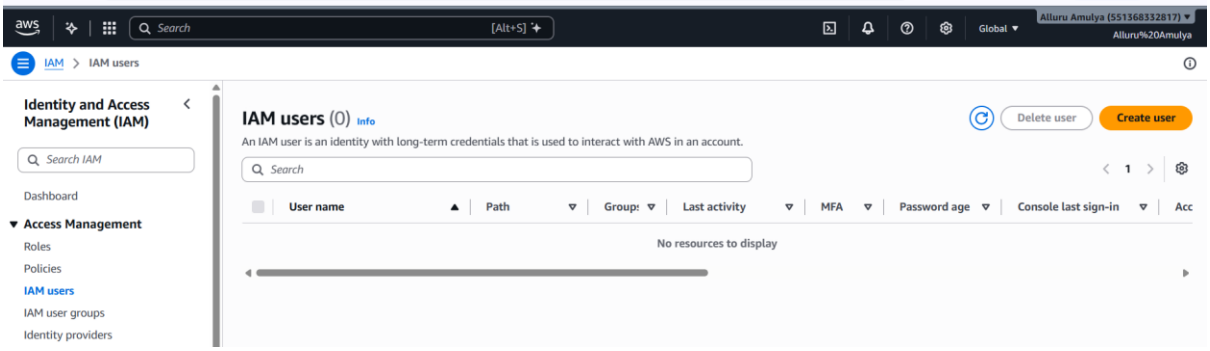
1. Open a web browser and navigate to the AWS Management Console.



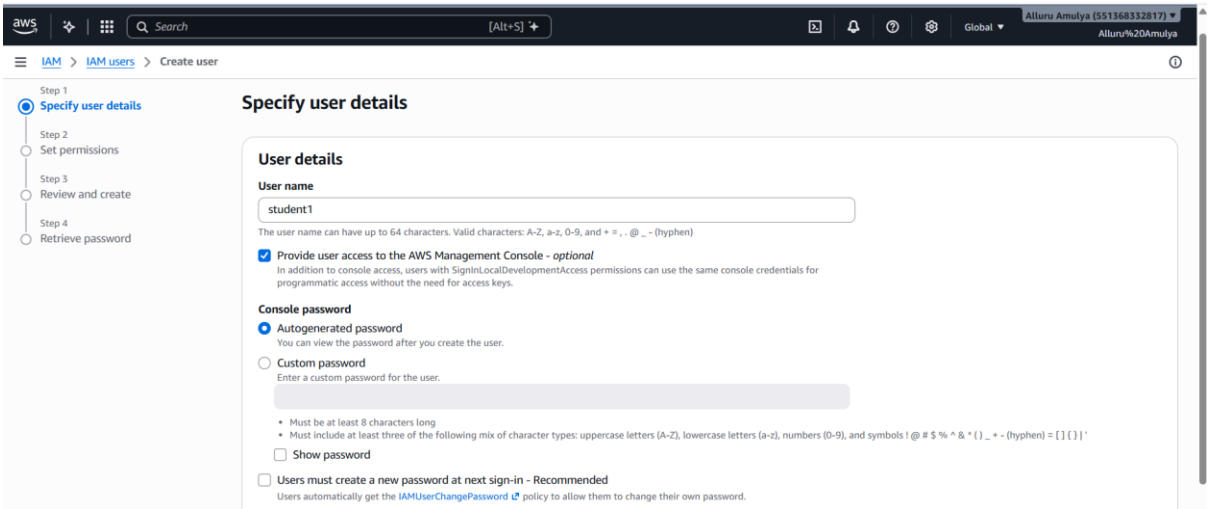
2. Sign in using the AWS root account credentials.
3. Search for IAM (Identity and Access Management) in the AWS Console search bar.



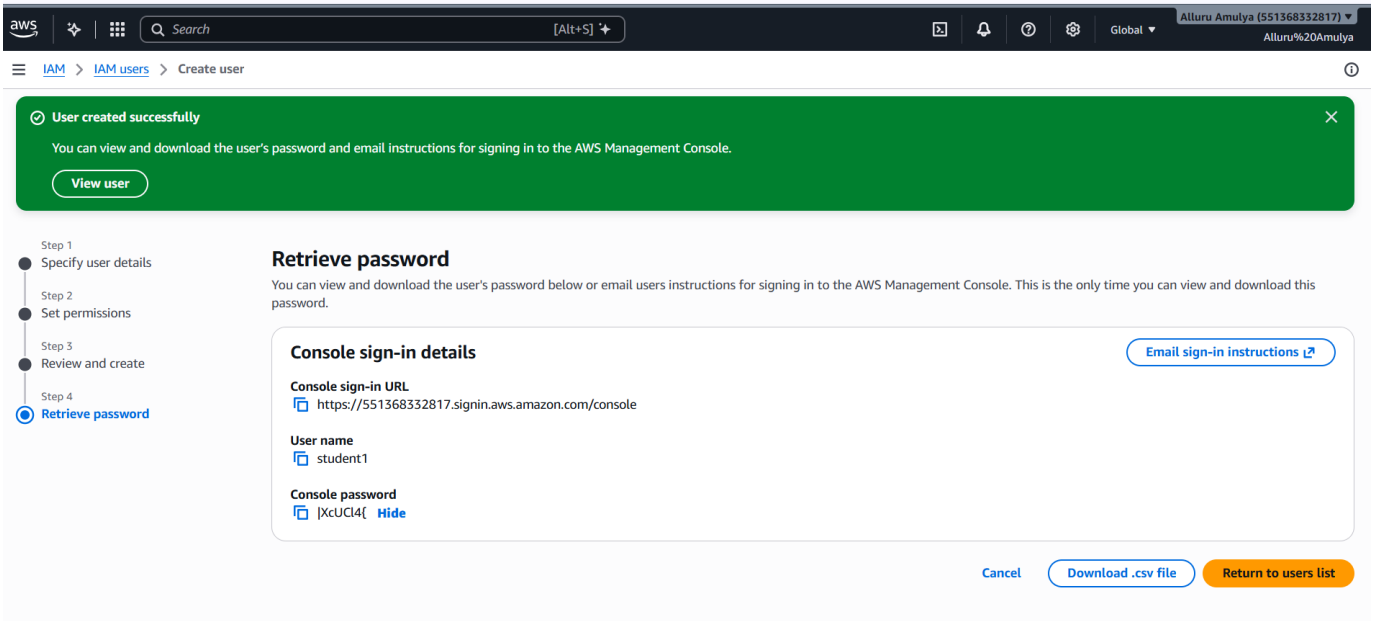
4. Select Users from the left navigation pane.



5. Click on Create user.
6. Enter a username (e.g., StudentUser01).
7. Enable console access for the user.



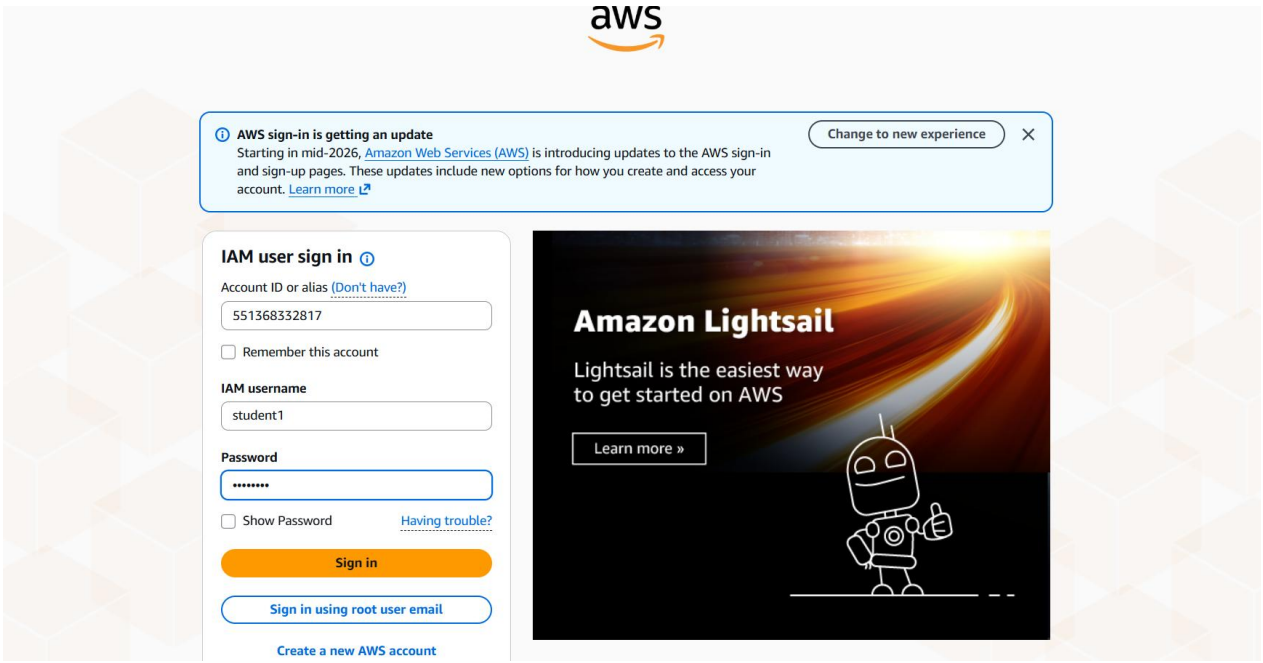
8. Set a custom password or select an autogenerated password.
9. Select the option to require a password change during the next sign-in, if required.
10. Under Set Permissions, select Attach policies directly.
11. Search for and select the AdministratorAccess policy.
12. Review the user details and permissions.
13. Click Create user to complete the process.
14. Save the IAM user sign-in details and password securely.



15. Sign out of the root account.

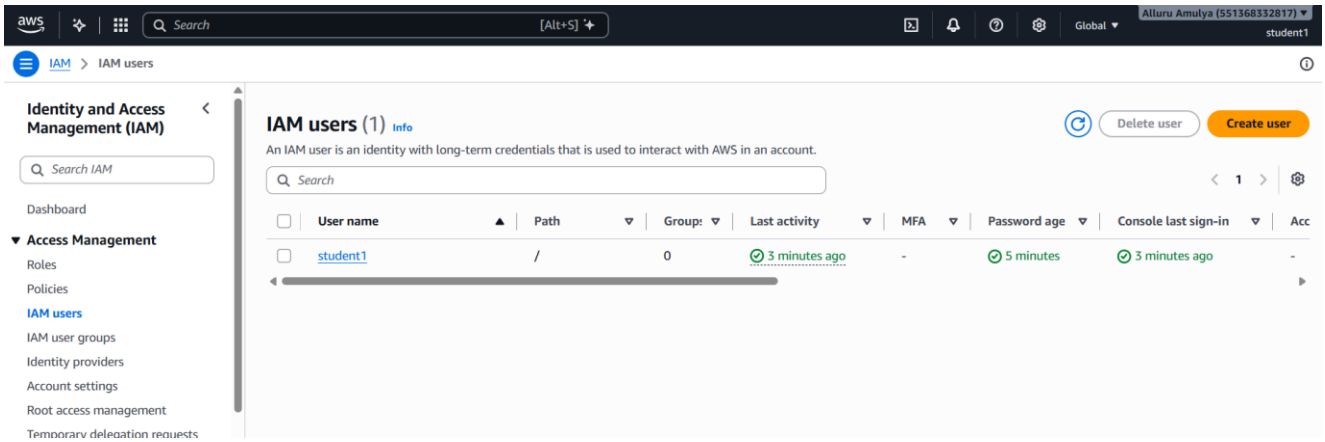
Part B: Signing in as an IAM User

1. Open the AWS Management Console sign-in page.
2. Select the IAM user sign-in option.
3. Enter the AWS account ID or account alias.
4. Enter the newly created IAM username and password.
5. If prompted, change the temporary password.
6. Verify that the IAM user can successfully access the AWS Management Console.

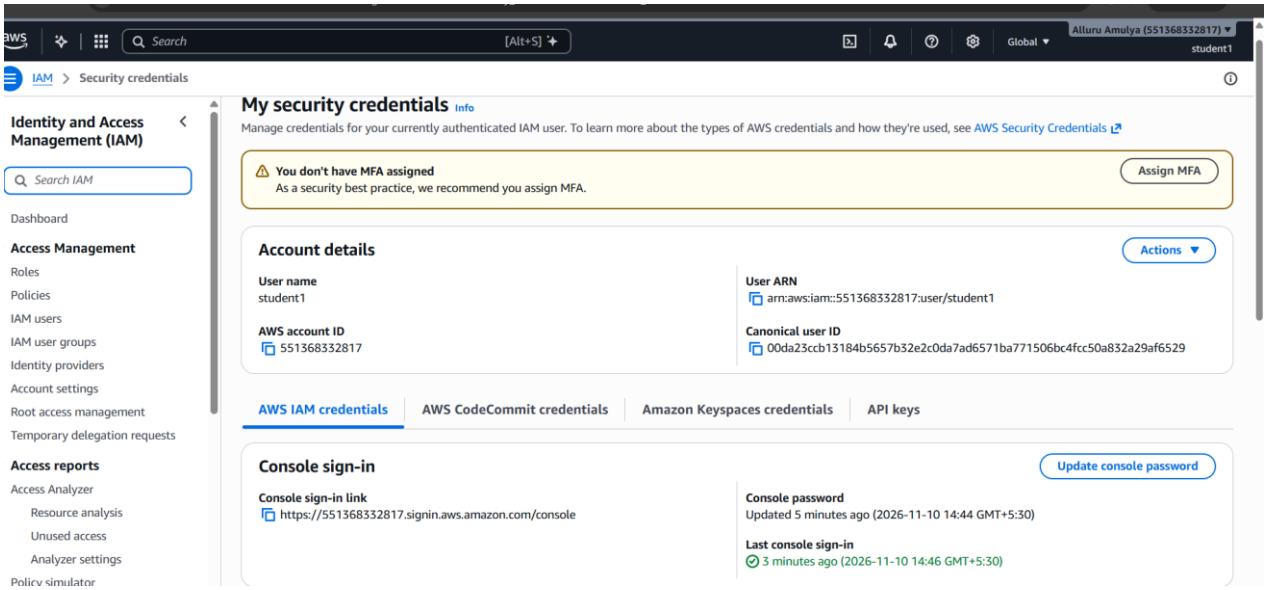


Part C: Enabling Multi-Factor Authentication (MFA)

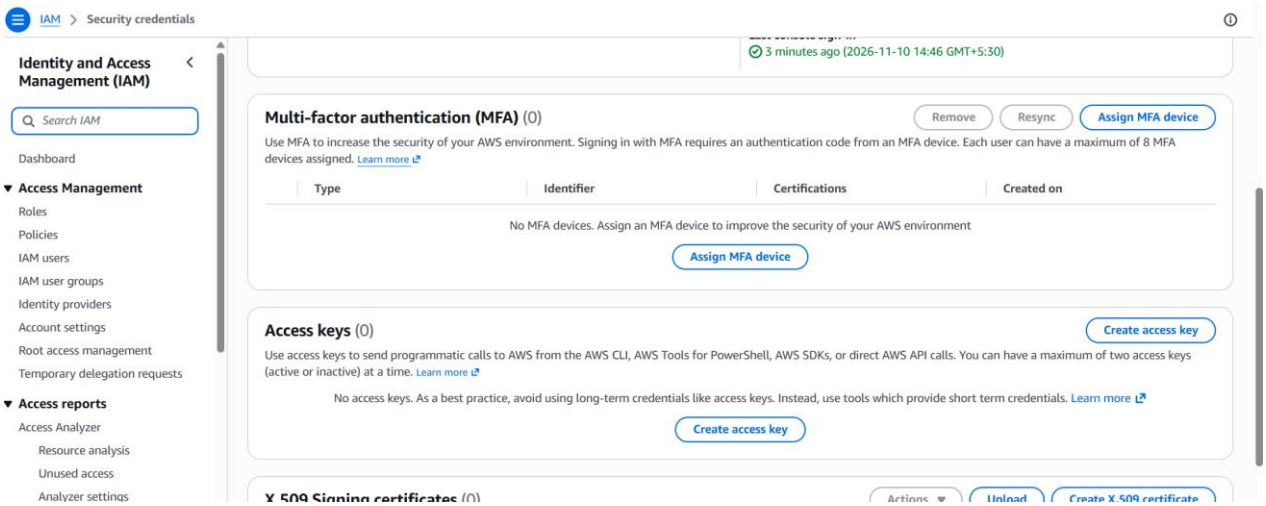
1. Sign in to the AWS Management Console using IAM user credentials.
2. Search for IAM and open the IAM dashboard.
3. Select Users from the left navigation pane.
4. Click on the newly created IAM username.



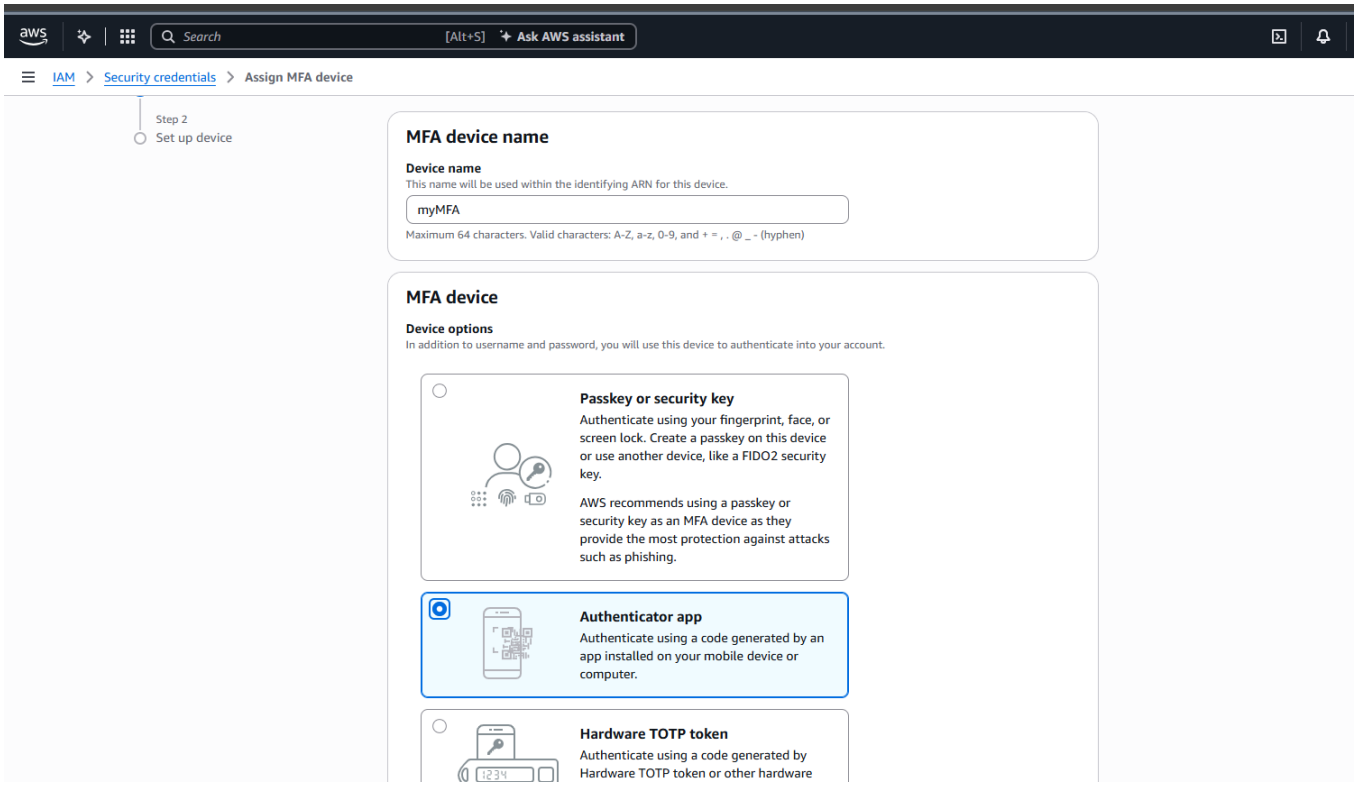
5. Open the Security credentials tab.



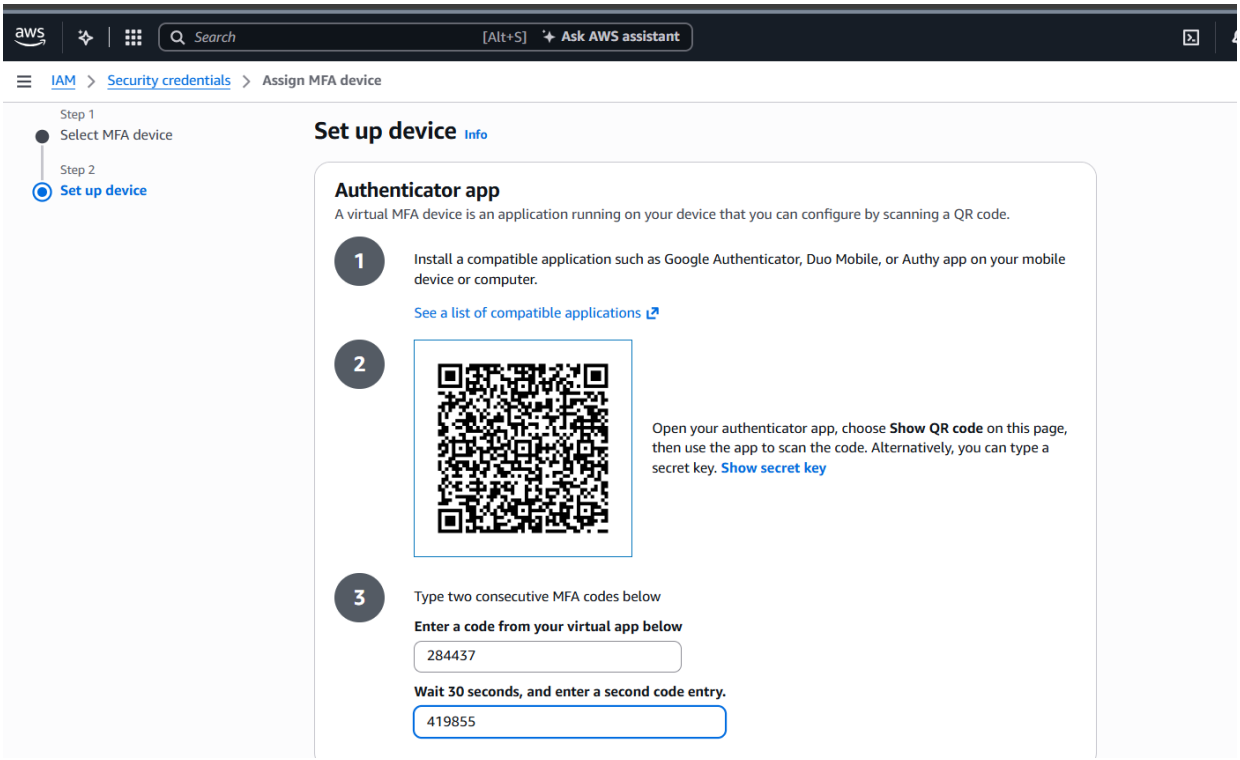
6. Locate the Multi-Factor Authentication (MFA) section.



7. Click on Assign MFA device.
8. Enter a device name (e.g., StudentMFA01).
9. Select an MFA device type:
    - Passkey or Security Key.
    - Authenticator Application (Google Authenticator or Microsoft Authenticator).

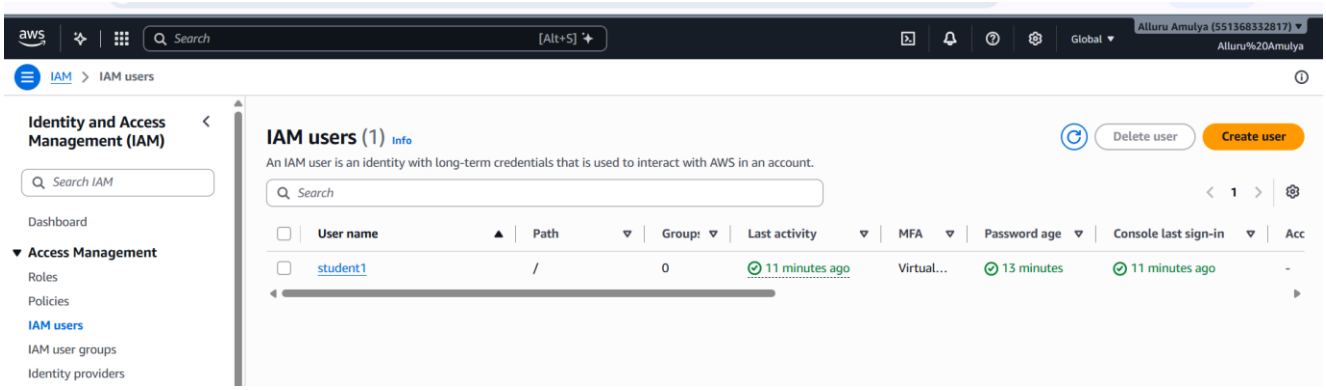


10. Follow the instructions displayed on the screen to register the selected MFA device.
11. If using an authenticator application, scan the QR code displayed in the AWS console.
12. Enter the first six-digit authentication code generated by the application.
13. Wait for the next authentication code and enter it.
14. Click Add MFA or Assign MFA to complete the setup.
15. Verify that the MFA device is successfully assigned to the IAM user.

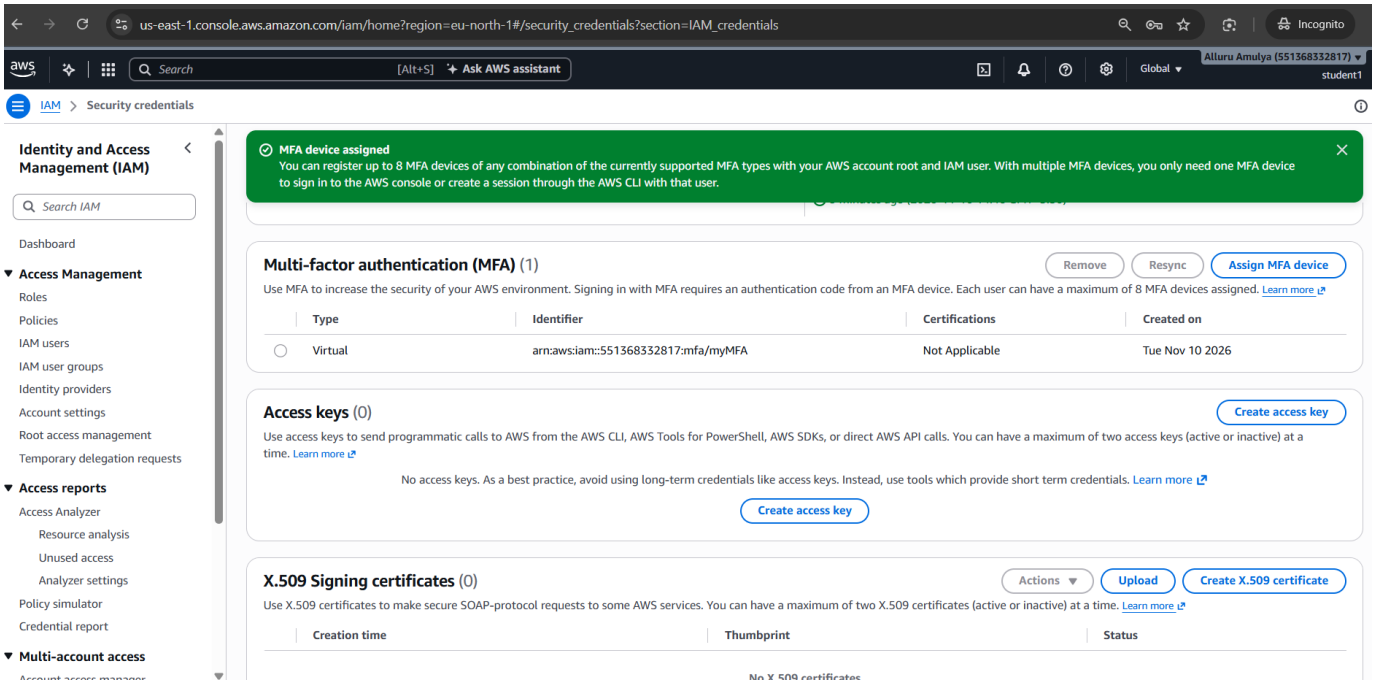


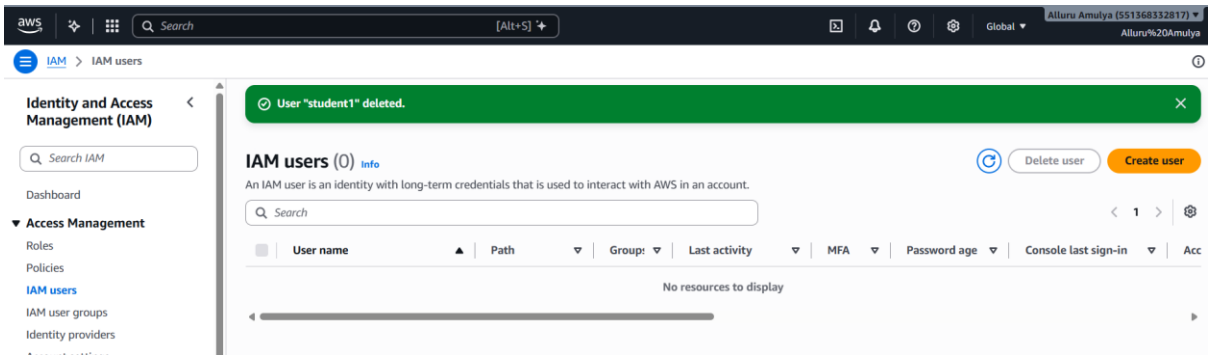
Part D: Testing MFA Authentication

1. Sign out of the AWS Management Console.
2. Open the AWS sign-in page.
3. Select IAM user sign-in.
4. Enter the AWS account ID or account alias.
5. Enter the IAM username and password.
6. When prompted, complete the MFA verification:
    - For an authenticator application, enter the current six-digit authentication code.
    - For a passkey or security key, complete verification using the registered device, PIN, or biometric authentication.
7. Verify that access to the AWS Management Console is granted successfully after MFA verification


Search
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Global
Alluru Amulya (551368332817)
Alluru%20Amulya
IAM > IAM users
Identity and Access Management (IAM)
Search IAM
Dashboard
Access Management
Roles
Policies
IAM users
IAM user groups
Identity providers
IAM users (1) Info
An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.
Delete user
Create user
Search
User name
Path
Group
Last activity
MFA
Password age
Console last sign-in
student1
/
0
11 minutes ago
Virtual...
13 minutes
11 minutes ago

Output:


us-east-1.console.aws.amazon.com/iam/home?region=eu-north-1#/security_credentials?section=IAM_credentials
Incognito
Search
[Alt+S]
Ask AWS assistant
Global
Alluru Amulya (551368332817)
student1
IAM > Security credentials
Identity and Access Management (IAM)
Search IAM
Dashboard
Access Management
Roles
Policies
IAM users
IAM user groups
Identity providers
Account settings
Root access management
Temporary delegation requests
Access reports
Access Analyzer
Resource analysis
Unused access
Analyzer settings
Policy simulator
Credential report
Multi-account access
MFA device assigned
You can register up to 8 MFA devices of any combination of the currently supported MFA types with your AWS account root and IAM user. With multiple MFA devices, you only need one MFA device to sign in to the AWS console or create a session through the AWS CLI with that user.
Multi-factor authentication (MFA) (1)
Remove
Resync
Assign MFA device
Use MFA to increase the security of your AWS environment. Signing in with MFA requires an authentication code from an MFA device. Each user can have a maximum of 8 MFA devices assigned. Learn more
Type
Identifier
Certifications
Created on
Virtual
arn:aws:iam::551368332817:mfa/myMFA
Not Applicable
Tue Nov 10 2026
Access keys (0)
Create access key
Use access keys to send programmatic calls to AWS from the AWS CLI, AWS Tools for PowerShell, AWS SDKs, or direct AWS API calls. You can have a maximum of two access keys (active or inactive) at a time. Learn more
No access keys. As a best practice, avoid using long-term credentials like access keys. Instead, use tools which provide short term credentials. Learn more
Create access key
X.509 Signing certificates (0)
Actions
Upload
Create X.509 certificate
Use X.509 certificates to make secure SOAP-protocol requests to some AWS services. You can have a maximum of two X.509 certificates (active or inactive) at a time. Learn more
Creation time
Thumbprint
Status


Search
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Global
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IAM > IAM users
Identity and Access Management (IAM)
Search IAM
Dashboard
Access Management
Roles
Policies
IAM users
IAM user groups
Identity providers
User "student1" deleted.
IAM users (0) Info
An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.
Delete user
Create user
Search
User name
Path
Group
Last activity
MFA
Password age
Console last sign-in
No resources to display